Preview: set-03/by_level/Grade_03-04/y2007_q21 (v1)

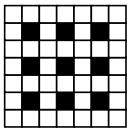
Growing patterns: describe how each diagram is built, then count cells = total – black

Part A: Each question is worth 3 points

1. These diagrams follow a pattern of black and white cells. If the pattern continues, how many white cells will the next diagram have?

8 white cells

21 white cells



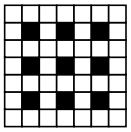
40 white cells

A. 69 ***B.*** 65 ***C.*** 64 ***D.*** 61 ***E.*** 59

2. These diagrams follow a pattern of black and white cells. If the pattern continues, how many white cells will the next diagram have?

8 white cells

21 white cells



40 white cells

A. 81 ***B.*** 65 ***C.*** 64 ***D.*** 61 ***E.*** 59

3. These diagrams follow a pattern: a black square with a white border around it. How many white cells will the next diagram have?

8 white cells

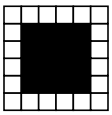
12 white cells

16 white cells

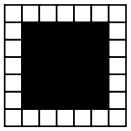
A. 16 ***B.*** 19 ***C.*** 20 ***D.*** 24 ***E.*** 36

4. These diagrams follow a pattern: a black square with a white border around it. How many white cells will the diagram after the next one have?

16 white cells



20 white cells



24 white cells

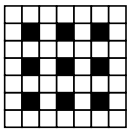
A. 49 ***B.*** 32 ***C.*** 31 ***D.*** 28 ***E.*** 24

Part B: Each question is worth 4 points

5. The diagrams below are the 1st, 2nd and 3rd of a pattern. How many white cells will the 6th diagram have?

8 white cells

21 white cells

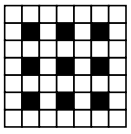

40 white cells

A. 96 ***B.*** 108 ***C.*** 114 ***D.*** 133 ***E.*** 169

6. The diagrams below are the 1st, 2nd and 3rd of a pattern. How many white cells will the 6th diagram have?

8 white cells

21 white cells

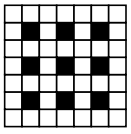

40 white cells

A. 176 ***B.*** 169 ***C.*** 166 ***D.*** 155 ***E.*** 133

7. The diagrams below are the 1st, 2nd and 3rd of a pattern. How many cells in all will the 4th diagram have?

8 white cells

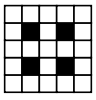
21 white cells

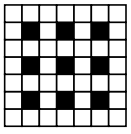

40 white cells

A. 16 ***B.*** 36 ***C.*** 64 ***D.*** 65 ***E.*** 81

8. The diagrams below are the 1st, 2nd and 3rd of a pattern. How many black cells will the 8th diagram have?

8 white cells

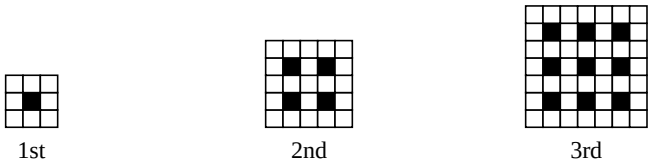

21 white cells


40 white cells

A. 64 ***B.*** 65 ***C.*** 68 ***D.*** 79 ***E.*** 81

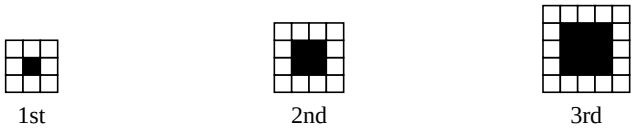
Part C: Each question is worth 5 points

9. The diagrams below are the 1st, 2nd and 3rd of a pattern. One diagram further on has exactly 96 white cells. Which one is it?



A. the 4th ***B.*** the 5th ***C.*** the 7th ***D.*** the 8th ***E.*** the 9th

10. The diagrams below are the 1st, 2nd and 3rd of a pattern. One diagram further on has exactly 40 white cells. Which one is it?



A. the 7th ***B.*** the 9th ***C.*** the 10th ***D.*** the 11th ***E.*** the 13th

Answer Key

1. **B**	2. **B**	3. **C**	4. **B**	5. **D**	6. **E**
7. **E**	8. **A**	9. **B**	10. **B**		

Solutions

1. B (65) — The sides grow by 2 and the black cells are 1 × 1, 2 × 2, 3 × 3, ... The next diagram is 9 × 9 = 81 cells with 4 × 4 = 16 black: 81 – 16 = 65. [set-03/by_level/Grade_03-04/2007 Q21 · next_white]

2. B (65) — The sides grow by 2 and the black cells are 1 × 1, 2 × 2, 3 × 3, ... The next diagram is 9 × 9 = 81 cells with 4 × 4 = 16 black: 81 – 16 = 65. [set-03/by_level/Grade_03-04/2007 Q21 · next_white]

3. C (20) — The border of an 6 × 6 square has 6 + 6 + 4 + 4 = 20 cells (or 6 × 6 – 4 × 4 = 20). [set-03/by_level/Grade_03-04/2007 Q21 · border_next]

4. B (32) — The border of an 9 × 9 square has 9 + 9 + 7 + 7 = 32 cells (or 9 × 9 – 7 × 7 = 32). [set-03/by_level/Grade_03-04/2007 Q21 · border_next]

5. D (133) — Diagram number k is (2k + 1) × (2k + 1) with k × k black cells. The 6th: 13 × 13 – 6 × 6 = 169 – 36 = 133. [set-03/by_level/Grade_03-04/2007 Q21 · far_diagram]

6. E (133) — Diagram number k is (2k + 1) × (2k + 1) with k × k black cells. The 6th: 13 × 13 – 6 × 6 = 169 – 36 = 133. [set-03/by_level/Grade_03-04/2007 Q21 · far_diagram]

7. E (81) — The 4th diagram is 9 × 9 = 81 cells, with 4 × 4 = 16 black ones. [set-03/by_level/Grade_03-04/2007 Q21 · black_and_total]

8. A (64) — The 8th diagram is 17 × 17 = 289 cells, with 8 × 8 = 64 black ones. [set-03/by_level/Grade_03-04/2007 Q21 · black_and_total]

9. B (the 5th) — Try the diagrams one by one: the 5th is 11 × 11 = 121 cells with 25 black, so 96 white. [set-03/by_level/Grade_03-04/2007 Q21 · which_diagram]

10. B (the 9th) — Try the diagrams one by one: the 9th is 11 × 11 = 121 cells with 81 black, so 40 white. [set-03/by_level/Grade_03-04/2007 Q21 · which_diagram]